

First Problem Sheet

on

Analytical Mechanics

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Topic: Newtonian Mechanics

“Millions saw the apple fall, but Newton was the one who asked why.”

-Bernard Baruch

Problem 1

Prove Proposition 1.1, 1.3 and 1.4.

Problem 2

Consider the transformation for rotating frames of reference $\mathbf{r}'(t) = \mathbf{R}(t)\mathbf{r}(t)$, $\mathbf{R}(t) \in SO(3)$.

- i) Prove that $\dot{\mathbf{R}}\mathbf{R}^T$ is antisymmetric and then show that any antisymmetric matrix $\Omega \in \mathcal{M}^{3 \times 3}$ can be written as $(\Omega)_{ij} = \varepsilon_{ijk}\omega_k$ for some $\boldsymbol{\omega} \in \mathbb{R}^3$.

Next, let $\mathcal{I} = \{O, e_1, e_2, e_3\}$ be an inertial reference frame, then consider the non-inertial reference frame $\mathcal{N} = \{O, e'_1, e'_2, e'_3\}$ with $e'_i = \mathbf{R}e_i$. Using direct differentiation, deduce:

- ii) The so-called **transport equation**

$$\frac{\mathcal{N}d}{dt}\mathbf{r}' = \frac{\mathcal{I}d}{dt}\mathbf{r}' + \boldsymbol{\omega} \times \mathbf{r}'.$$

- iii) The analogue of Newton's second law of motion for rotating (noninertial) frames

$$\mathcal{N}\mathbf{F} = \mathcal{I}\mathbf{F} + 2\boldsymbol{\omega} \times \mathcal{I}\mathbf{v}' + \dot{\boldsymbol{\omega}} \times \mathbf{r}' + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}').$$

Further, identify each term in the latter expression.

Problem 3

Prove Theorem 1.7 (Conservation Theorem for the Angular Momentum of a Particle).

Problem 4

Prove that if U is a simply connected open subset of $\mathbb{R}^n$, then the following statements are equivalent:

- i) $\int_{\Gamma} \mathbf{F} = \int_{\Sigma} \mathbf{F}$ for all contours $\Gamma, \Sigma \subset U$ joining the same initial and final points.
ii) $\int_D \mathbf{F}$ for all closed contours $D \subset U$.
iii) $\mathbf{F} = \nabla V$ for some $V : U \rightarrow \mathbb{R}$ at least class $C^1(U)$.

Moreover, show that for the case $n = 3$ we also have the equivalence with

- iv) $\nabla \times \mathbf{F} = 0$ on U .

Hint: for iv) you may want to look up Stoke's theorem from vector analysis.

Problem 5

Show that every central force is conservative. In particular, for Newton's gravitational force, find the potential $V(r)$.

Problem 6

If a projectile is fired from the origin of the coordinate system with an initial velocity v_0 and in a direction making an angle α with the horizontal, calculate the time required for the projectile to cross a line passing through the origin and making an angle $\beta < \alpha$ with the horizontal (neglect the effects of drag forces).

Problem 7

Consider a particle undergoing horizontal motion in a medium having a drag force satisfying Stoke's law. Find the position and velocity of the particle. Show that the particle eventually stops (as expected) and find the total range. If the particle started with a velocity $\mathbf{v}_0 = v_0 \hat{\mathbf{x}}$, find the work done by the drag force on the trajectory of the particle.

Problem 8

Consider a particle vertically falling in a medium satisfying Stoke's law. Find the position and velocity of the particle. Furthermore, compute

i) $\lim_{t \rightarrow \infty} \mathbf{v}(t),$

ii) $\lim_{t \rightarrow \infty} \mathbf{r}(t),$

iii) W_d (the work done by the drag force)

and discuss whether the results are physically consistent.

Problem 9

Consider the motion of a projectile fired with a velocity $\mathbf{v}_0$. Neglect the effects of the air resistance and thus find its position, velocity and range. If air resistance is taken into consideration (assume that Stoke's law holds), calculate the decrease in range by using the **perturbation method**.

Problem 10

Consider a particle of mass m whose motion starts from rest in a constant gravitational field. If a drag force satisfying Stoke's law is encountered, show that the distance s the particle falls in accelerating from v_0 to v_1 is given by

$$\ln \left(\frac{g - kv_1}{g - kv_0} \right) + (v_0 - v_1).$$

Furthermore, prove that if a drag force proportional to the square of the velocity is encountered, then s is given by

$$d = \frac{1}{2k} \ln \frac{g - kv_0^2}{g - kv_1^2}$$

where k is the proportionality factor.

Problem 11

A particle moves in a medium under the influence of a retarding force equal to $mk(v^3 + a^2v)$, where k and a are constants. Show that for any value of the initial speed the particle will never move a distance greater than $\pi/2ka$ and that the particle comes to rest only for $t \rightarrow \infty$.

Problem 12

Consider a particle moving in the region $x > 0$ under the influence of the potential

$$V(x) = V_0 \left(\frac{a}{x} + \frac{x}{a} \right)$$

where $V_0 = 1J$ and $a = 2m$. Plot the potential, find the equilibrium points, and determine whether they are maxima or minima.

Problem 13

Write the criteria for determining whether an equilibrium is stable or unstable when all derivatives up through order n vanish. Prove your assertion.

Problem 14

Consider a particle of mass m under the influence of the potential

$$V(\mathbf{r}) = V_0 \left[\left(\frac{x}{a} \right)^2 + \left(\frac{y}{a} \right)^2 - \frac{z/a}{1 + (z/a)^2} \right]$$

with V_0, a positive constants.

- i) Find the force $\mathbf{F}$ acting on the particle.
- ii) Find all the equilibrium points and determine whether they are stable, unstable or metastable (saddle point).
- iii) Sketch or describe distinct level surfaces of V .